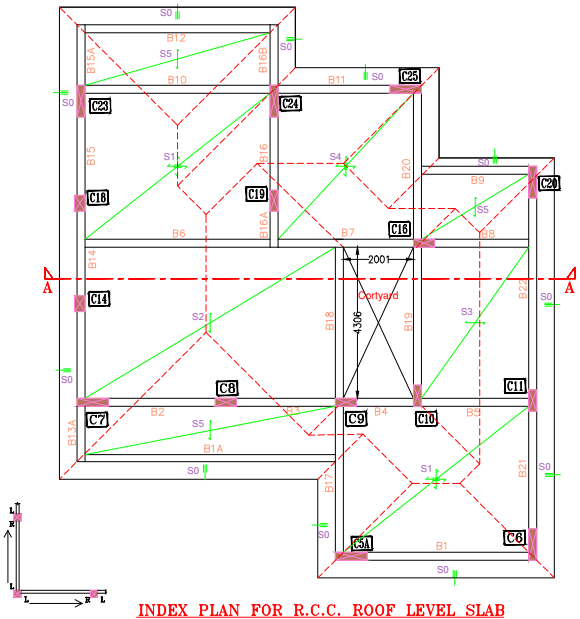
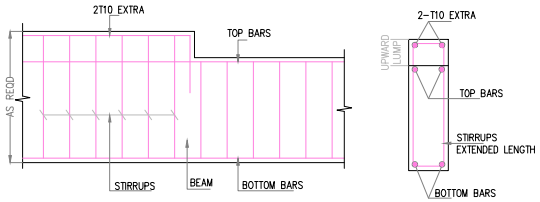
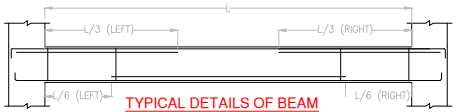


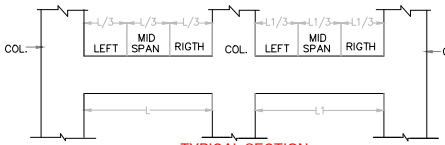
INDEX PLAN FOR R.C.C. ROOF LEVEL SLAB



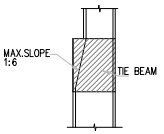
DETAILS OF UPWARD LUMP (TYPICAL)



TYPICAL DETAILS OF BEAM



TYPICAL SECTION
(INDICATES LEFT, MID AND RIGHT SPAN OF BEAM)



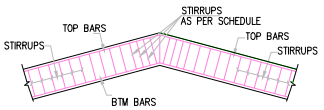
AT ANY LVL WHERE COLUMN SIZE
GETS REDUCED IN EITHER DIMENSION
BEAM IS ABSOLUTELY ESSENTIAL

BEAM SCHEDULE

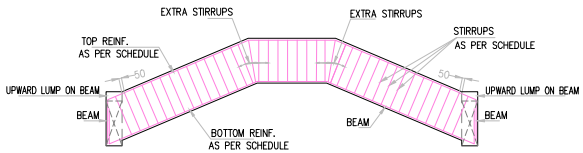
BEAM NUMBERS	SIZE		BOTTOM REINFORCEMENT			TOP REINFORCEMENT			SHEAR STIRRUPS			REMARKS
	B	D	LEFT	MID SPAN	RIGHT	LEFT	MID SPAN	RIGHT	LEFT	MID SPAN	RIGHT	
B1	230	600	3-T16	+ 3-T16 2-T16	3-T16	2-T10	2-T10	2-T10	2L-T8@150 C/C	2L-T8@150 C/C	2L-T8@150 C/C	-
B1A	200	600	2-T16	+ 2-T16 2-T12	2-T16	2-T12	2-T12	2-T12	2L-T8@150 C/C	2L-T8@150 C/C	2L-T8@150 C/C	-
B2	230	600	2-T16	+ 2-T16 2-T16	2-T16	2-T10	2-T10	2-T10	2L-T8@150 C/C	2L-T8@150 C/C	2L-T8@150 C/C	CRANKED BEAM
B3	230	600	2-T16	+ 2-T16 1-T16	2-T16	2-T10	2-T10	2-T10	2L-T8@200 C/C	2L-T8@200 C/C	2L-T8@200 C/C	CRANKED BEAM
B4	230	600	+ 2-T12 1-T10	+ 2-T12 1-T10	+ 2-T12 1-T10	2-T10	2-T10	2-T10	2L-T8@250 C/C	2L-T8@250 C/C	2L-T8@250 C/C	-
B5	230	600	3-T12	3-T12	3-T12	2-T10	2-T10	2-T10	2L-T8@250 C/C	2L-T8@250 C/C	2L-T8@250 C/C	CRANKED BEAM
B6	230	600	3-T20	+ 3-T20 2-T20	3-T20	2-T16	2-T16	2-T16	2L-T10@150 C/C	2L-T10@150 C/C	2L-T10@150 C/C	CRANKED BEAM
B7	230	600	2-T16	+ 2-T16 2-T16	2-T16	2-T10	2-T10	2-T10	2L-T8@200 C/C	2L-T8@200 C/C	2L-T8@200 C/C	-
B8	230	600	3-T12	3-T12	3-T12	2-T10	2-T10	2-T10	2L-T8@250 C/C	2L-T8@250 C/C	2L-T8@250 C/C	CRANKED BEAM
B9	230	600	+ 2-T12 1-T10	+ 2-T12 1-T10	+ 2-T12 1-T10	2-T10	2-T10	2-T10	2L-T8@250 C/C	2L-T8@250 C/C	2L-T8@250 C/C	-
B10	230	600	3-T16	+ 3-T16 2-T16	3-T16	2-T10	2-T10	2-T10	2L-T8@150 C/C	2L-T8@200 C/C	2L-T8@150 C/C	CRANKED BEAM
B11	230	600	3-T12	3-T12	3-T12	2-T10	2-T10	2-T10	2L-T8@250 C/C	2L-T8@250 C/C	2L-T8@250 C/C	-
B12	230	450/D	3-T12	+ 3-T12 1-T12	3-T12	2-T10	2-T10	2-T10	2L-T8@250 C/C	2L-T8@250 C/C	2L-T8@250 C/C	AS PER SLAB PROFILE
B13+B13A	230	600	2-T16	2-T16	2-T16	+ 2-T16 2-T16	+ 2-T16 2-T16	+ 2-T16 2-T16	2L-T8@150 C/C	2L-T8@150 C/C	2L-T8@150 C/C	BARS IN ONE PIECE
B14	230	600	3-T20	+ 3-T20 2-T12	3-T20	2-T12	2-T12	2-T12	2L-T8@150 C/C	2L-T8@150 C/C	2L-T8@150 C/C	-
B15+B15A	230	450/D	2-T16	2-T16	2-T16	+ 2-T16 2-T12	+ 2-T16 2-T12	+ 2-T16 2-T12	2L-T8@125 C/C	2L-T8@125 C/C	2L-T8@125 C/C	BARS IN ONE PIECE AS PER SLAB PROFILE
B16A	230	600	3-T25	3-T25	3-T25	+ 3-T25 2-T25	+ 3-T25 2-T25	+ 3-T25 2-T25	2L-T10@100 C/C	2L-T10@100 C/C	2L-T10@100 C/C	CANTILEVER (SPAN-1035 MM)
B16	230	450/D	2-T16	2-T16	2-T16	2-T25	2-T25	2-T25	2L-T8@125 C/C	2L-T8@125 C/C	2L-T8@125 C/C	CRANKED BEAM
B16B	230	450/D	2-T16	2-T16	2-T16	2-T25	2-T25	2-T25	2L-T8@125 C/C	2L-T8@125 C/C	2L-T8@125 C/C	AS PER SLAB PROFILE
B17	230	600	3-T16	+ 3-T16 2-T12	3-T16	2-T12	2-T12	2-T12	2L-T8@150 C/C	2L-T8@150 C/C	2L-T8@150 C/C	CRANKED BEAM
B18	230	600	3-T12	3-T12	3-T12	1-T10	1-T10	1-T10	2L-T8@250 C/C	2L-T8@250 C/C	2L-T8@250 C/C	-
B19	230	600	+ 2-T12 1-T10	+ 2-T12 1-T10	+ 2-T12 1-T10	2-T10	2-T10	2-T10	2L-T8@250 C/C	2L-T8@250 C/C	2L-T8@250 C/C	-
B20	230	600	2-T16	+ 2-T16 2-T16	2-T16	2-T10	2-T10	2-T10	2L-T8@200 C/C	2L-T8@200 C/C	2L-T8@200 C/C	CRANKED BEAM
B21	230	600	3-T12	+ 3-T12 1-T12	3-T12	2-T10	2-T10	2-T10	2L-T8@250 C/C	2L-T8@250 C/C	2L-T8@250 C/C	-
B22	230	600	2-T20	+ 2-T20 2-T20	2-T20	2-T12	2-T12	2-T12	2L-T8@150 C/C	2L-T8@150 C/C	2L-T8@150 C/C	PROVIDE U/W LUMP

SCHEDULE OF RCC SLABS

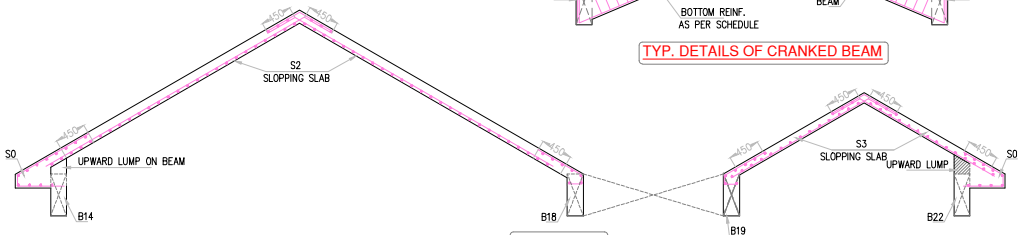
TYPE	THICKNESS	STEEL ALONG SPAN ↕	STEEL ACROSS SPAN ↔	REMARKS
S1	165	T10 @ 165 C/C ALT BENT UP	T10 @ 175 C/C ALT BENT UP	-
S2	165	T10 @ 125 C/C ALT BENT UP	T8 @ 200 C/C DIST	-
S3	125	T8 @ 125 C/C ALT BENT UP	T8 @ 230 C/C DIST	-
S4	150	T8 @ 140 C/C ALT BENT UP	T8 @ 150 C/C ALT BENT UP	-
S5	115	T8 @ 150 C/C ALT BENT UP	T8 @ 230 C/C DIST	-
S0	150	T10 @ 100 C/C ALL STR @ TOP EVERY ALT BAR BENT BACK ALONG BOTTOM	T8 @ 200 C/C DIST AT TOP & BOTTOM	CANTILEVER SLAB



TYPICAL DETAILS OF CRANKED BEAM



TYP. DETAILS OF CRANKED BEAM



SECTION:A-A

SIGNATURE OF ARCHITECT ON RECORD:-

SIGNATURE & STAMP OF JW CONSULTANTS LLP:-

DRAWING IS VALID FOR DESIGN ADEQUACY, PROVIDED IT IS SIGNED & STAMPED BY OUR OFFICE

NOTES:

1. Basic reference codes:
   IS456: 2000, IS1893(Part1): 2002, IS13920: 1993(Reaffirmed 2002)
2. Due care shall be taken to ascertain that requisite strength of concrete is gained before commencement of deshuttering. It shall comply with provisions of Clause No. 11.3 of IS 456: 2000.
3. Nominal covers

		Mild	Moderate	Severe
I	Footings	50	50	50
II	Columns & walls >200mm width (to links of column)	40	40	45
III	Columns & walls with width of 200mm & below having reinf. of dia. 16mm & above. (to links of column)	40	40	45
IV	Columns & walls with width of 200mm & below having reinf. of dia. 12mm. (to links of column)	25	30	45
V	Slabs	20	30	45
VI	Beams (to stirrups of beam)	20	30	45
VII	Lift walls	40	40	45

   For main reinf. up to 12mm dia. bar for mild exposure the nominal cover may be reduced by 5mm for slabs & beams only.
4. Beams with depth more than 750mm, provide side-face reinforcement.
5. Substratum shall be approved from our office before laying P.C.C.
6. Minimum clear spacing between any two longitudinal bars in beam= 50mm.
7. All laps (Ld) shall be staggered & not more than 50% bars to be lapped at any given section.

GRADE OF REINF.	M20	M25	M30	M35	M40 & ABOVE
Fe415	48 X D	41 X D	38 X D	34 X D	30 X D
Fe500 (TMT)	57 X D	49 X D	46 X D	40 X D	36 X D

   For bundled bars, Ld shall be increased by 10% for 2 bars in contact, 20% for 3 bars in contact and 33% for 4 bars in contact.
8. All buildings shall have tie beams/plinth beams at ground/plinth level.
9. If footings overlap with each other, necessary revision shall be obtained from our office.
10. Design is valid for number of floors as indicated in the drawing.
11. At any level where column size gets reduced in either dimension tie beams/plinth beams are essential.
12. For cantilevers, top bars to be anchored behind from external face of support for - Ld or span of cantilever - whichever is greater.
13. Fire rating considered:- 1 Hour max.

Use of this drawing for construction shall explicitly confirm acceptance of following conditions by Owner / Builder / Contractor
1. Our responsibility shall remain limited to safe and sound structural design as transmitted by this drawing.
2. This drawing shall not be used for any construction unless necessary statutory approvals are obtained by the Owner / Builder / Architect from the Regulating Authorities. JW CONSULTANTS LLP are indemnified of any legal consequences in case of unauthorised construction.
3. We shall not remain responsible for -
   a) Safety of old structure during demolition.
   b) Safety of any adjoining building /persons staying in adjoining building/persons and properties on adjoining roads.
   c) Safety of construction worker/any personnel at work site during construction
   d) Correctness/safety of any temporary structure, scaffolding, shuttering, centering erected at site and any injury to any personnel arising out of any accidents.
   e) Accidents occurring due to premature deshuttering, faulty / substandard construction material or workmanship / faulty construction procedure.
   f) Any accident occurring due to construction of elements of buildings not designed by us.
4. Supervision if specifically asked for will be provided to the extent of verification of reinforcement on site but responsibility regarding correct & sound construction shall solely rest with contractor/ builder / owner.
5. All structural concrete should be weigh batched, machine mixed and mechanically vibrated.
6. Any discrepancy between our drawing & Architects' drawing shall be brought to our notice before construction.
7. This drawing is property of JW CONSULTANTS LLP and shall not be reproduced or used without explicit permission of this office.

DRAWING TO BE VALIDATED BY ARCHITECT ON RECORD AFTER CHECKING COMPLIANCE WITH APPROVED PLANS FROM THE REGULATING LOCAL GOVERNMENT AUTHORITY.

NO.	REV. DATE	DESCRIPTION	NO.	REV. DATE	DESCRIPTION
R0	17.10.2022	INITIAL DRAWING	R5	.	.
R1		.	R6	.	.
R2	.	.	R7	.	.
R3	.	.	R8	.	.
R4	.	.	R9	.	.

ENVIRONMENTAL EXPOSURE CONDITION :-	MILD	GRADE OF CONCRETE :-	M25
DESIGN LIVE LOAD :- (UNLESS SPECIFIED)	-	GRADE OF STEEL :-	Fe 500 TMT
SUPERIMPOSED DEAD LOAD :-	-	S.B.C. :-	40T/M2

CHECKED BY	DEALT BY	DRAWN BY	DRG. NO.	BLDG:	JOB NO.
A.U.P.	N.C.P.	M.S.J.	080024-R10A-411	BUNGALOW-213+214	080024

DEVELOPER:-	- MATRIX DEVELOPER PVT LTD.
DESIGN ARCHITECT:-	- A.B. VAIDYA ASSOCIATES
ARCHITECT ON RECORD:-	- A.B. VAIDYA ASSOCIATES
PROJECT:-	- R10A BHILARE BUNGLOWS AT "FOREST TRAILS" BHUGAON
TITLE:-	- INDEX PLAN ,SCHEDULE FOR R.C.C. ROOF FLOOR SLAB & BEAMS

LLP IDENTITY NO.: AAA-2650
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STRUCTURAL ENGINEERS
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